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Batch: A-2

Division: A

Subject: Programming with Java

ASSIGNMENT No : 7

TITLE : Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives

To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp7/Main1.java
This animal eats food.
Dog barks!
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp7/Main2.java
Name: Alice
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Exercise

Q1. Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp7/Exercise/Ex1.java
Created a new shape!
Area of Circle : 314.0
Created a new shape!
Area of Rectangle : 100.0
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Develop an E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp7/Exercise/Ex2.java
Product ID : 101
Product Name : Laptop
Price : 65000.0
Category : Electronic

Product ID : 102
Product Name : T-Shirt
Price : 999.0
Category : Clothing

Product ID : 103
Product Name : Rice
Price : 1200.0
Category : Grocery
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Conclusion

This assignment helped in understanding the concepts of inheritance and interfaces in Java and their role in object-oriented programming. It demonstrated how inheritance promotes code reusability by allowing child classes to inherit the properties and methods of a parent class, while interfaces provide abstraction by defining a common set of behaviors that different classes can implement. The implementation improved my understanding of polymorphism, modular programming, and software design principles. These concepts are widely used in developing desktop applications, web applications, Android applications, enterprise software, gaming systems, and other real-world Java applications that require reusable, flexible, and maintainable code.